

ORDER-DEPENDENT NONLINEAR CLOSED-ORBIT RESPONSE IN A CESR MODEL

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Abstract

DRAFT PLACEHOLDER. Write after the mixed horizontal-vertical second-order experiment and the sextupole/octupole attribution scan are complete.

INTRODUCTION

DRAFT PLACEHOLDER. Add the motivation, the relation to linear orbit-response matrices and nonlinear closed-orbit methods, and a concise statement of the paper's contribution after the result set is frozen.

ORBIT AND OPTICS CALCULATION

A Bmad-compatible model of the Cornell Electron Storage Ring (CESR) was used with 119 corrector variables: 58 horizontal and 61 vertical correctors. The orbit benchmark contains 1000 deterministic input samples, including the zero-control reference. The nonzero inputs are independently sampled Gaussian corrector changes with an rms value of $5\text{ }\mu\text{rad}$, clipped at three standard deviations. Closed orbits are recorded at 99 zero-length detector markers in both transverse planes.

SciBmad solves the radio-frequency-on closed orbits as a batch. A cached 6×119 nominal closed-orbit response matrix supplies the initial estimate $z_0 + R\Delta k$; Newton iterations reuse the nominal Jacobian and send failed lanes to a full automatic-differentiation fallback. All 1000 formal samples converged without fallback. On the common 1nx201 host, the maintained SciBmad run required 33.178 s, compared with 67.370 s for Bmad/Tao. The 198000 compared detector coordinates had correlation 0.999999966 and a maximum absolute difference of $8.14\text{ }\mu\text{m}$. The latter includes differences between the independently represented lattices and is not a Newton closure tolerance.

The associated radio-frequency-off optics benchmark evaluates periodic coupled Twiss quantities and their first momentum derivatives at the same 99 markers. Table 1 summarizes representative formal results. The parameterized calculation uses one third-order phase-space map with 119 first-order corrector parameters; it is a local surrogate rather than 1000 independent optics solutions.

Table 1: Representative 1000-Sample Calculation Results

Calculation	Time [s]	Rate [s ⁻¹]	Median corr.
Bmad RF-on orbit	67.370	14.843	1.000000
SciBmad RF-on orbit	33.178	30.140	1.000000
SciBmad point optics	570.771	1.752	1.000000
Parameterized optics	73.130	13.674	0.999996

ORBIT RESPONSE ERROR ANALYSIS

For corrector perturbation Δk , the exact detector orbit is expanded about the nominal machine as

$$u(\Delta k) = u_0 + J\Delta k + \frac{1}{2}H[\Delta k, \Delta k] + \frac{1}{6}T[\Delta k, \Delta k, \Delta k] + \dots \quad (1)$$

The residual of the nominal orbit-response approximation is $e = u(\Delta k) - u_0 - J\Delta k$. Inputs are written as $\Delta k = \rho d$ ($5\text{ }\mu\text{rad}$), where every active part of d has unit rms. A sweep over 600 fixed Gaussian directions compares all-corrector, horizontal-only, and vertical-only inputs. It contains 36001 nonlinear states: one reference plus 20 radii, 600 directions, and three input scenarios. Every state converged through $\rho = 12.8$.

The mean detector root-mean-square error $E(\rho)$ is calculated separately for the horizontal and vertical detector coordinates. Quadratic response is diagnosed by constant E/ρ^2 and a local log-log slope near two. The small-radius coefficients in Table 2 show two strong anisotropies. First, the all-corrector horizontal residual is close to the vertical-only value. Second, the all-corrector vertical coefficient is about 73 times the vertical-only coefficient and 145 times the horizontal-only coefficient. The latter cannot be reproduced by either isolated plane and therefore motivates a direct mixed-Hessian test.

Table 2: Mean Quadratic Coefficients at $\rho = 0.1$

Active correctors	x residual	y residual
	[$\mu\text{m}/\rho^2$]	
All	0.5648	0.5321
Horizontal only	0.1570	0.003660
Vertical only	0.5524	0.007270

The all-corrector channels remain near quadratic through the complete common range. Horizontal-only remains near quadratic through $\rho = 64$. The clear exception is the vertical detector residual for vertical-only inputs: it leaves the quadratic guide around $\rho = 0.28$ – 0.4 , and its slope approaches three while the full nonlinear reference remains converged.

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ALL-CORRECTOR SECOND-ORDER SOURCE

DRAFT PLACEHOLDER. Insert the four-sign ($\pm h, \pm v$) experiment and the measured decomposition into H_{hh} , H_{hv} , and H_{vv} . The planned test must show whether the large all-corrector vertical residual is reproduced by $H_{y,hv}[h, v]$ and quantify the reconstruction error. Do not state the mixed-term attribution as established until these data exist.

VERTICAL-CORRECTOR SIGNED PARITY

For a midplane-symmetric ideal lattice, reversal of the vertical-corrector vector gives

$$x(v) = x(-v), \quad y(v) = -y(-v). \quad (2)$$

Consequently, the vertical detector response to vertical-only input contains odd powers of v . After subtraction of the linear response, its leading allowed nonlinear term is cubic. By contrast, the horizontal detector response permits a quadratic v^2 term. This is the closed-orbit form of the normal-multipole midplane selection rule [1].

The order was tested directly with 100 fixed 61-dimensional vertical-corrector directions and 27 radii. Both signs of every direction were evaluated, giving 5401 converged nonlinear states with no fallback and a maximum one-turn closure norm of 9.994×10^{-14} . For detector plane u , the paired components are

$$e_{\text{even}} = \frac{u(+\Delta k) + u(-\Delta k)}{2} - u_0, \quad (3)$$

$$e_{\text{odd, nl}} = \frac{u(+\Delta k) - u(-\Delta k)}{2} - J\Delta k. \quad (4)$$

Symmetric positive and negative corrector excitations are also the standard central-difference construction used for measured orbit-response matrices [2].

The vertical even component follows ρ^2 , whereas the odd nonlinear component follows ρ^3 . Their mean detector rms values cross at $\rho \simeq 1.31$, corresponding to an active-corrector rms kick of about $6.5 \mu\text{rad}$. At $\rho = 6.4$, the mean odd/even ratio is 4.91. The horizontal detector control remains overwhelmingly even: its odd/even ratio is 3.05×10^{-4} at the same radius.

The transition is direction dependent. At $\rho = 6.4$, the median odd/even ratio is 2.48 and 80% of directions are odd dominated. At the extension point $\rho = 10.05$, the 10th, 50th, and 90th percentiles of f are 50.06%, 93.84%, and 99.76%, respectively. Thus at least 90% of the sampled directions are odd dominated at that amplitude.

Multipole attribution

DRAFT PLACEHOLDER. Insert the normal-sextupole and octupole ablation/strength scans. Recompute the linear response for every modified lattice and fit the direction-resolved cubic response vector, rather than comparing only one large-amplitude rms value. Test a sextupole-cascade scaling with K_2^2 against a direct octupole scaling with K_3 , including interference if both are significant.

IMPLICATIONS FOR RESPONSE SURROGATES

Equation (1) shows why one global truncation order is not sufficient. A quadratic Hessian is appropriate for channels whose residuals retain ρ^2 scaling over the intended input distribution. It cannot represent the vertical-only vertical response after the odd cubic component becomes comparable to the symmetry-breaking even term. A practical surrogate should therefore select order by input/output channel, retain exact nonlinear solutions as validation references, and report its validity radius together with direction-dependent error statistics. Nonlinear closed-orbit and higher-order response measurements have similarly been used to identify sextupole behavior and other nonlinear lattice information [3–5].

FUTURE OPTICS ERROR ANALYSIS

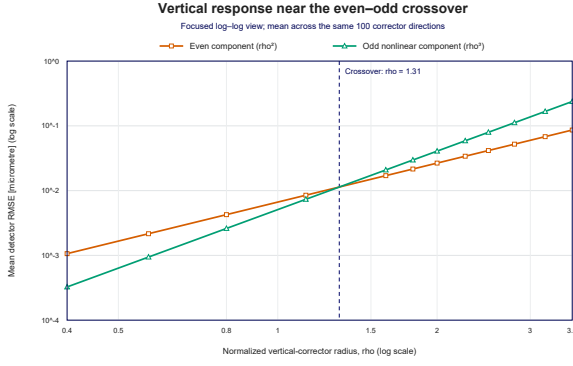
The same order-versus-amplitude analysis will be applied to ordinary optics, chromatic quantities, coupling observables, and orbit-derived fields. These classes will be reported separately because aggregation can hide a small quadratic coefficient or an early higher-order transition. This extension is future work and is not used to support the orbit conclusions above.

CONCLUSION

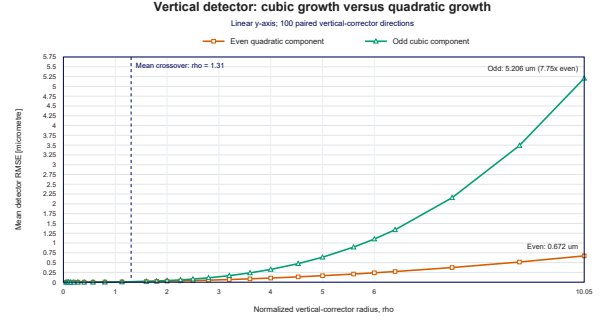
The CESR-model benchmarks establish reproducible closed-orbit and optics calculations on a shared 1000-sample input set. The orbit-radius sweep then identifies a channel-specific failure of a purely quadratic residual model. Signed-direction measurements show that the vertical-only vertical residual is an odd cubic response overtaking a small even component, rather than a generic numerical loss of convergence. The two remaining causal tests are a mixed-Hessian decomposition of the all-corrector vertical residual and a sextupole/octupole attribution of the cubic coefficient.

REFERENCES

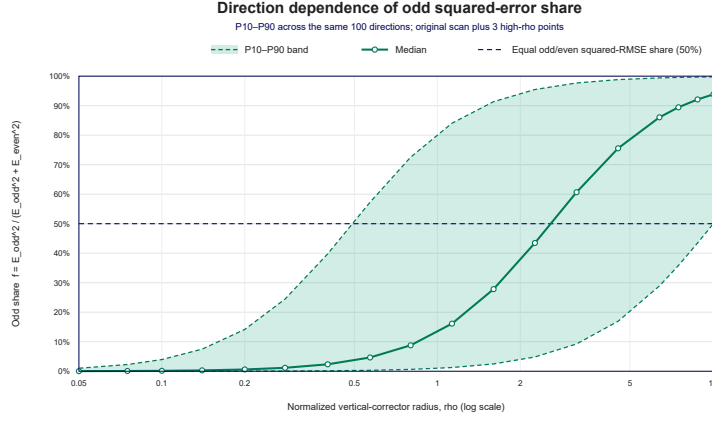
- [1] A. Wolski, “Maxwell’s equations for magnets”, in *Proc. CAS 2009: Magnets*, Bruges, Belgium, Jun. 2009, pp. 1–38. doi:10.5170/CERN-2010-004.1
- [2] J.-P. Thiry, A. Dieckmann, F. Frommberger, W. Hillert, and J. F. Schmidt, “Lattice matching with Elegant at ELSA”, in *Proc. IPAC’16*, Busan, Korea, May 2016, pp. 2690–2692. doi:10.18429/JACoW-IPAC2016-WEPOR011



(a) Focused log-log crossover.



(b) Full-range linear-scale growth.



(c) Direction-dependent odd squared-error share.

Figure 1: Signed-parity analysis of the vertical detector residual. The quadratic even and cubic odd nonlinear components cross near $\rho = 1.31$. For each direction, the bounded odd share is $f = E_{\text{odd}}^2 / (E_{\text{odd}}^2 + E_{\text{even}}^2)$; the band reports its 10th, 50th, and 90th percentiles over the same 100 directions.

- [3] D. K. Olsson, A. Andersson, and M. Sjöström, “Nonlinear optics from off-energy closed orbits”, *Phys. Rev. Accel. Beams*, vol. 23, p. 102803, 2020.
doi:10.1103/PhysRevAccelBeams.23.102803
- [4] X. Huang, “Correction of nonlinear lattice with closed orbit modulation”, in *Proc. IPAC’24*, Nashville, TN, USA, May 2024, pp. 3054–3056.
doi:10.18429/JACoW-IPAC2024-THPC32

- [5] C. Caliri, A. Oeftiger, and O. Boine-Frankenheim, “Beam-based identification of magnetic field errors in a synchrotron using deep Lie map networks”, *Phys. Rev. Accel. Beams*, vol. 28, p. 024601, 2025.
doi:10.1103/PhysRevAccelBeams.28.024601